National

# Premise Health Test (PHT)

HOW11339 | MODIFIED 09/15/2025



## Summary

Learn more about PHT.

## Quick Links


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- MoCA PHY rates
- MoCA (Segmented network, bridged network, and expected devices)
- FM ingress
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- PHT Pause (Test throttling)
- Unresponsive device
- Unrecognizable device
- Incomplete D3P data
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- Missing PHY rate message — Call from X-Ray failed to return PHY rates for all devices
- PHT analysis types and descriptions — These analysis are given when data suggests one of these conditions could exist
- Filter — Possible noise filter blocking one of the upstreams from connecting
- Outage — All devices offline and matching outage in NWT
- POE filter — When another device is detected inside a MoCA mesh network that is not on the account (neighbor device likely)
- SNR Delta — Delta between Port SNR at CMTS and cable modem SNR
- ALL OUT — All devices offline and unresponsive; No matching NWT event or outage

- PHT failures and National Watchtower (NWT) node event override
- Troubleshooting
- PHT failures and NWT node event override


## What's changed in this article

| Date | Summary of change |
|---|---|
| September 15, 2025 | Article maintenance — no content updates. |
| September 11, 2025 | Article review. |
| September 22, 2023 | Overhaul. Extensive reorganization and edits. |

## Overview

PHT is a tool to assist technicians in the diagnosis of any premise. It is an enhancement over Home Integrity Check (HIC). HIC validates only DOCSIS parameters of a device while PHT can validate DOCSIS and video devices (devices must be full band capture capable).

It includes:

- Being able to see the full band spectrum of devices with capable chipsets.
- Multimedia over Coax Alliance (MoCA) scoring.

- In-depth FLUX/in-channel frequency response (ICFR) graphs to help with impairments found in the network within the home.
- Orthogonal Frequency-Division Multiplexing (OFDM) and OFDMA measurements.
- The ability to link to National Watchtower (NWT) and Problem Manager to see if there is an open ticket or network event for the customer.

## PHT metrics and thresholds

| Metric | Upper fail | Lower fail | Single threshold fail |
|---|---|---|---|
| Actual upstream (US) TX | > 54 dBmV | < 25 dBmV | |
| US TX vBng EPON | > -4.0 dBmV | < -28 dBmV | |
| Partial bonding | *Mid-split capable devices only | | Registration state < > 4<br>Downstream (DS) state < > 1<br>*Orthogonal frequency-division multiple access (OFDMA) ranging status < > 4 |

| Metric | Upper fail | Lower fail | Single threshold fail |
|---|---|---|---|
| FLUX ICFR | | | > = 3 dB ICFR (ICFR indicators) |
| DS RX | > 13 dBmV | < -13 dBmV | |
| DS RX EPON | > -10 dMmV | < -28 dBmV | See market Refer to Maintenance (RTM) process for any failing light level conditions |
| DS (OFDM) RX | > 15 dBmV | < -15 dBmV* | |
| DS SNR | | | < 33 dB |
| DS OFDM mean MER (SNR) | | | < 33 dB |
| Spectrum impairments* | *Full spectrum devices only | | Any individual impairment < > Adjacent Channel Power |

| Metric | Upper fail | Lower fail | Single threshold fail |
|---|---|---|---|
| MoCA PHY rate*<br>MoCA network* | *MoCA capable devices only<br>*MoCA segmented network devices<br>*MoCA unexpected and/or foreign devices | | < 200 Mbps (XG to XI, XB to Xi, XG to XB devices only; Xi to Xi or RNG150 to RNG150 not in scope) |
| FM ingress | *All DOCSIS devices | | Severe ingress condition |
| All out | | | Two or more DOCSIS devices, all are known by CMTS, but offline and unresponsive |
| WiFi–RSSI | *Re-launch 10-29-19; None Pass/Fail | | < -70 dBm RSSI range at Xi5/Xi6 |

**Notes:**

- All Upstream (US) and Downstream (DS) channels will be tested for each device on the account.
- Any US or DS channel failure for any metric on any device will result in test failure.
- Spectrum Full Band Capture (FBC) impairments and/or subpar MoCA PHY rates on any device will result in test failure.

- Active DOCSIS devices on a customer's account that are offline will result in a test failure if the device has been active on a cable modem termination system (CMTS) within the last 24 hours.
  - **Note:** This will **fail on new installs** within the first three days.
    - We should not leave active devices offline on a new install.
  - If the device is offline longer than 24 hours, the result would be a warning and the device would be considered unrecognizable.
- All failures **must be fixed** before leaving the customer premise (obvious reasons why we cannot pass would include RTM (overrides), construction, access issues).
- Open NWT node events will be cross referenced to override test failures when applicable.
- OFDM values apply to DOCSIS 3.1 devices.
- OFDMA values apply to DOCSIS 3.1 devices.
- PHT results fall into three possible results categories when Test Profile is **tech**:
  - **Pass (pure pass):** All devices in the home are passing all threshold specifications.
  - **Pass with overrides (Pass w/ Overrides):** Items outside the home, which the technician has no control over.
  - **Fail:** Items are potentially within the technician's control; failure is not associated with any other events.

## Actual US TX

The maximum US TX level for a DOCSIS device depends on how many US channels the device is using for US bonding. A growing number of devices in our network have Upstream Extended Transmit (UET) capabilities and can transmit 3 dBmV higher than the initial DOCSIS 3.0 specification allowed.

| Device capability | Modulation scheme | Modulation profile | Maximum transit level per channel | | |
|---|---|---|---|---|---|
| | | | One channel | Two channels | Three or four channels |
| No UET | 64 QAM | A-TDMA | 57 dBmV | 54 dBmV | 51 dBmV |
| With UET | 64 QAM | A-TDMA | 60 dBmV | 57 dBmV | 54 dBmV |
| **UET allows an extra 3 dBmV of TX power** | | | | | |

If a device is transmitting at maximum power and still cannot reach the CMTS at the requested input level, generally 0 dBmV, it is in a state called "compression." Compression limits the device's equalization abilities and forces constant power adjustment requests from the CMTS. This is where "Actual US TX" comes in.

**Note:** Actual US TX is the level a device is transmitting at, or would need to transmit at, in order to reach the CMTS at the proper input level. If a device is not in compression, the Actual US TX is the current transmit level. When a device is in compression, the Actual US TX is calculated as the current transmit level plus the absolute value of the CMTS RX Level for that channel.

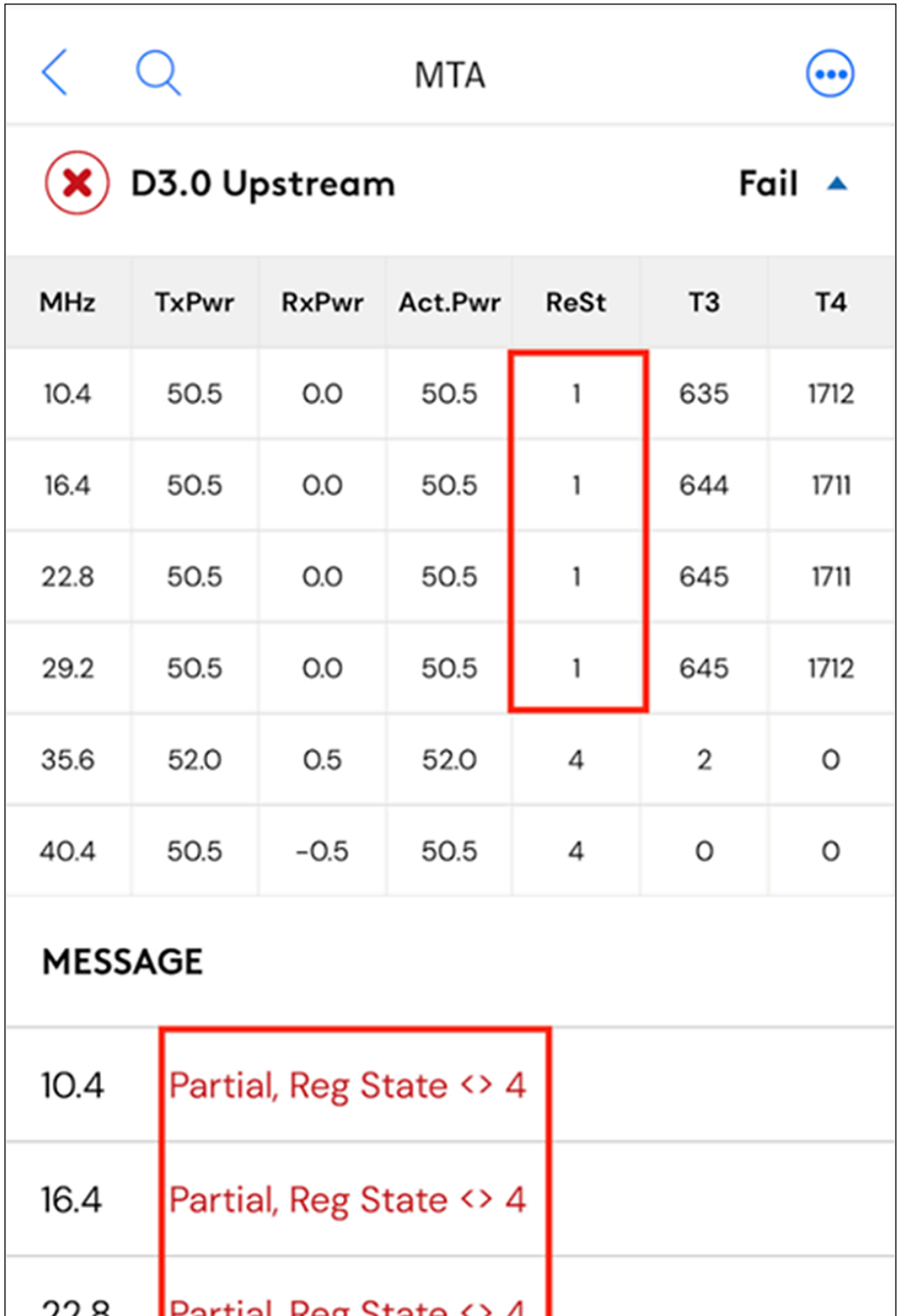
E.g., If a device is transmitting at 51 dBmV and the CMTS RX is -3.5 dBmV, then the Actual US TX will be 51 + 3.5 = 54.5 dBmV. This results in a failed test.

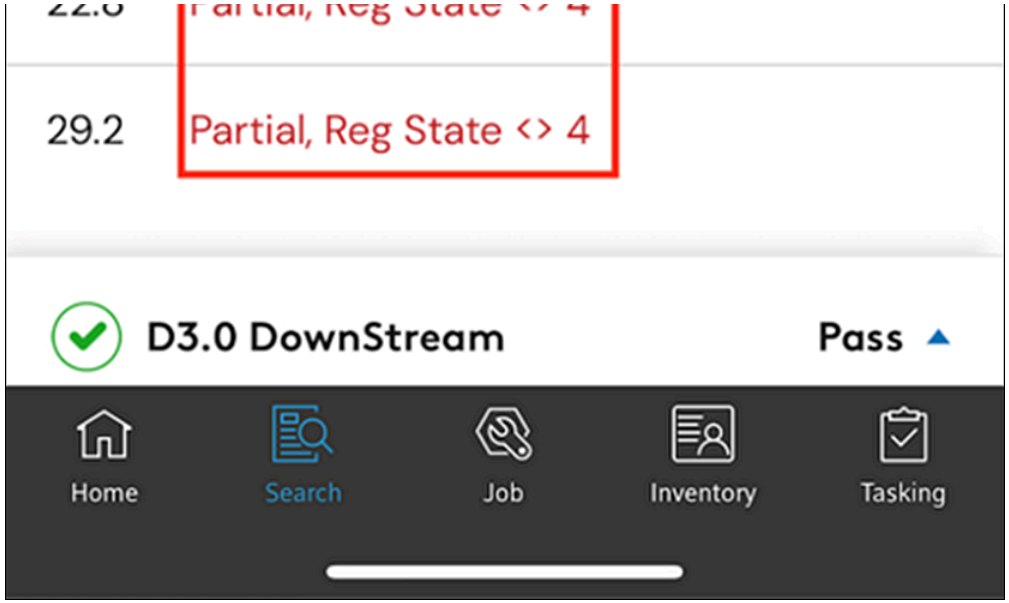
| No UET | | | |
|---|---|---|---|
| Frequency | TX level | RX level | Actual TX |
| 17.3 MHz | 51 | 0 | 51 |
| 23.7 MHz | 47 | 0 | 47 |
| 30.1 MHz | 51 | -2.1 | 53.1 |
| 36.5 MHz | 51 | -5.7 | **56.7** |

| With UET | | | |
|---|---|---|---|
| Frequency | TX level | RX level | Actual TX |
| 17.3 MHz | 52.8 | 0 | 52.8 |
| 23.7 MHz | 54 | -1 | **55** |
| 30.1 MHz | 47.3 | 0 | 47.3 |
| 36.5 MHz | 52.3 | 0 | 52.3 |

## Partial bonding

DOCSIS 3.0 devices have the ability to send and receive data transmissions across multiple US and DS channels; this is called "bonding." If any US or DS channel(s) becomes impaired, the device will stop using them to send or receive data. This is called "partial bonding." If you have a partial bonding condition with the device, PHT will trigger a failing condition.


MTA
D3.0 Upstream
Fail
MHz TxPwr RxPwr Act.Pwr ReSt T3 T4
10.4 50.5 0.0 50.5 1 635 1712
16.4 50.5 0.0 50.5 1 644 1711
22.8 50.5 0.0 50.5 1 645 1711
29.2 50.5 0.0 50.5 1 645 1712
35.6 52.0 0.5 52.0 4 2 0
40.4 50.5 -0.5 50.5 4 0 0
MESSAGE
10.4 Partial, Reg State <> 4
16.4 Partial, Reg State <> 4

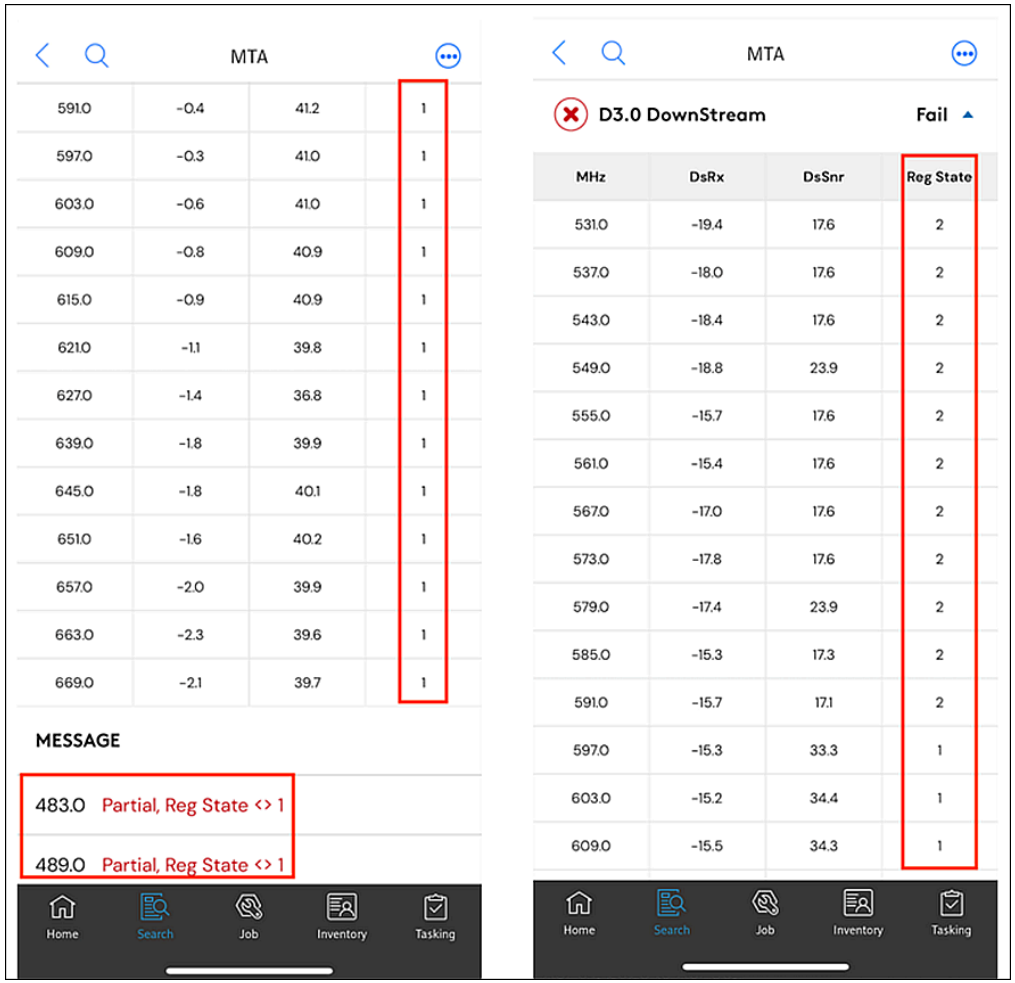



Devices running in partial bonding mode indicate current or past impairments on the channel(s) that are no longer being used. Customers with devices in partial bonding mode may not get the speeds they are subscribed for and may also contribute to node level capacity issues affecting speeds for the entire node.

Partial mode devices are identified by the DOCSIS 3.0 registration state for each channel. A registration state of 4 on any channel indicates that the device is using that channel for data transmissions. The values are very similar for DOCSIS 3.1.

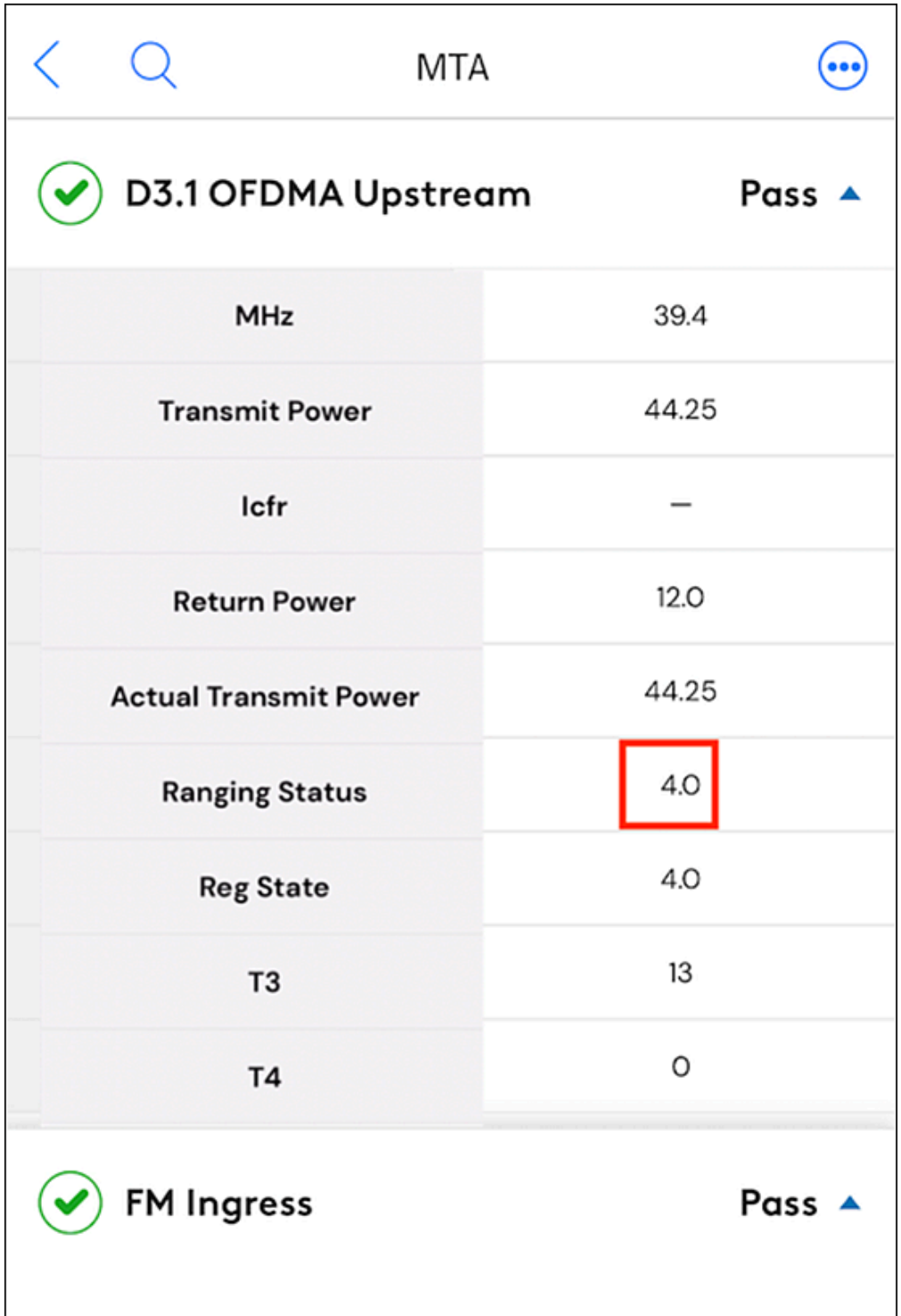
For DS OFDM, the registration state of 4 indicates that the device is using that channel for data transmissions.

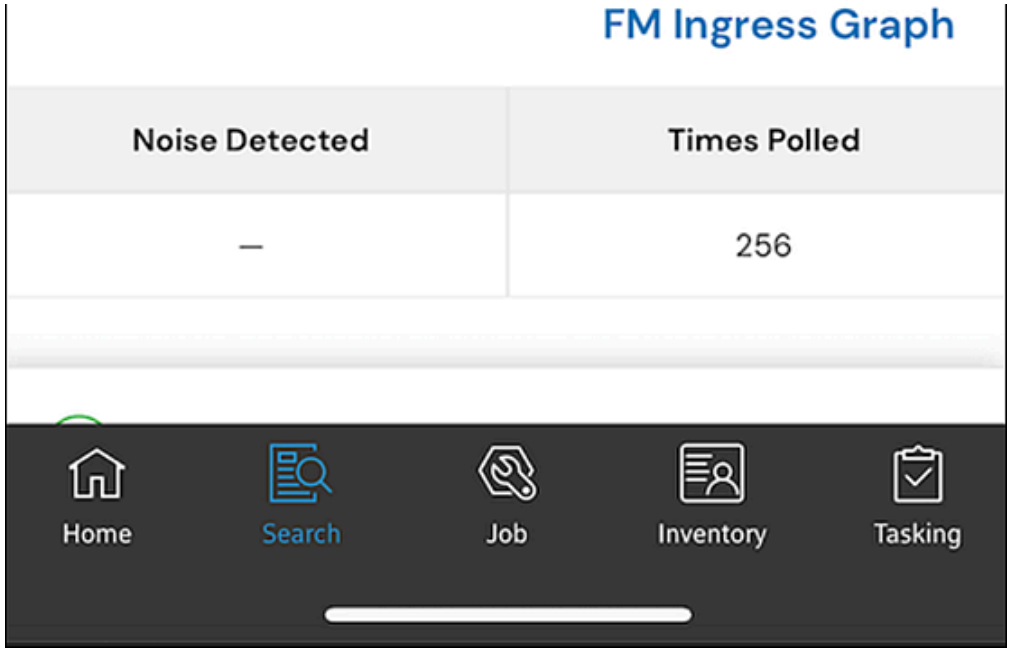
For US OFDMA, a ranging status of 4 is the equivalent. A registration state of 1, 2, 3, or 5 on any channel indicates that a device is not using the channel(s) and is in partial bonding mode.




**Note:** Use the XM OFDMA widget on your meter to validate OFDMA bonding and health at the tap and CPE: 


Successful response for **OFDMA Upstream:**


MTA
D3.1 OFDMA Upstream
Pass
MHz
39.4
Transmit Power
44.25
Icfr
–
Return Power
12.0
Actual Transmit Power
44.25
Ranging Status
4.0
Reg State
4.0
T3
13
T4
0
FM Ingress
Pass




## DS RX and DS SNR

DS RX and DS SNR are key metrics used to determine the health of our DOCSIS carriers at the input of our devices. Low levels in either category can result in performance issues that impact our customers.

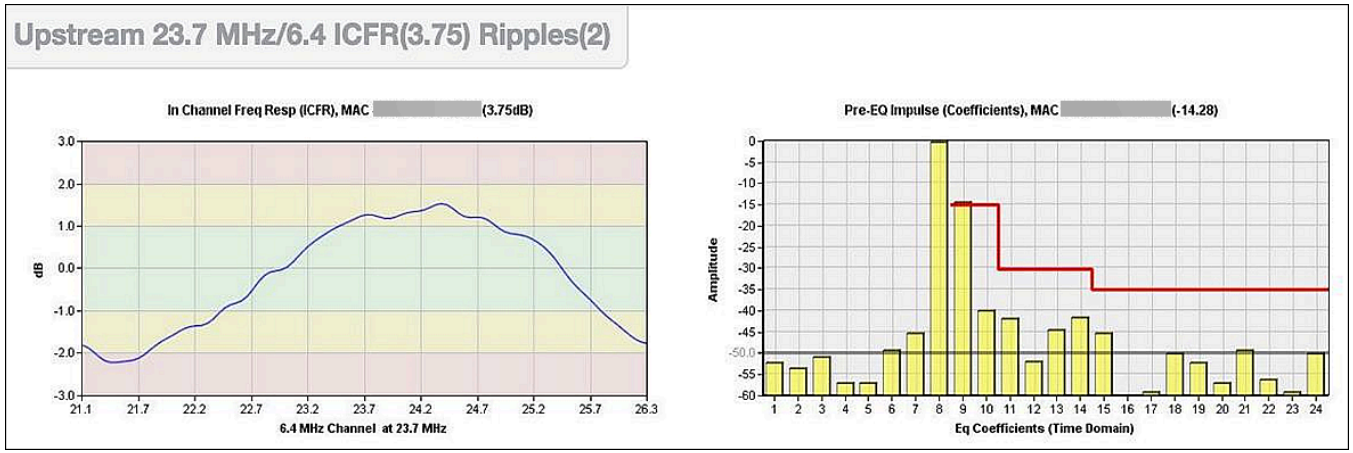
However, low DS SNR is often found to be related to low DS RX levels though they can occur separately. While there is no upper failure limit for DS SNR, high DS RX levels can also be harmful to our devices. If DS RX levels are too high, it can overload the front end of the device and cause performance issues.

## FLUX ICFR

The ICFR of our US QAMs gives us insight into impairments in the US path that we are often otherwise blind to. We can delineate between customer premise and HFC network related issues.

Devices that display ICFR issues on any US channel that are localized to the customer premise indicate integrity problems within the home and/or drop cabling. Things as simple as loose connectors can result in ICFR issues and, if left unchecked, loose connections can become an ingress and/or noise source affecting the performance of not only that individual customer, but the whole node area.

Example of a high ICFR due to a premise issue resulting in a test failure:




## ICFR Flux indicators — v5 PHT release/enhancement

This enhanced logic within PHT has been designed to help us narrow in on the source of an ICFR signature impairment affecting a device's ICFR signature. The updated logic takes several device comparisons and correlates the devices within neighboring area to make a cleaner decision on whether the ICFR signature impairment is specific to the account a technician is working on or part of a larger network issue.

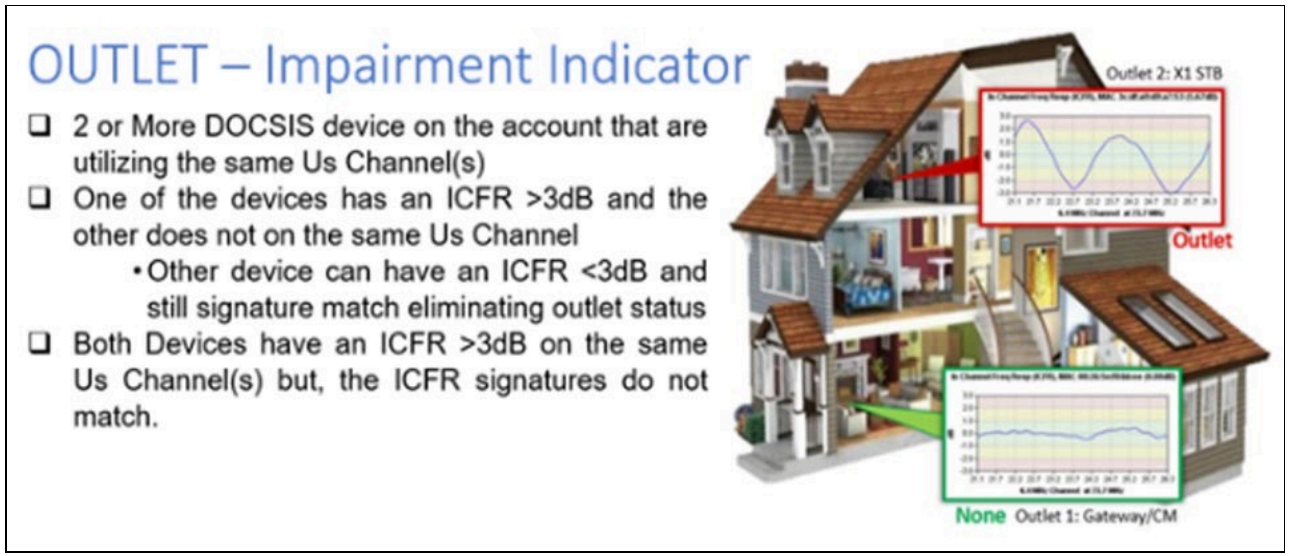
After some additional testing and conversations with each of the key stakeholders, we have vetted out the logic to ensure all five ICFR Impairment Indicators (Solo, Outlet, Home, Drop, and Network).

Technicians will need to be conscientious and ensure there are No Potential Loose Connectors indicators initiated. NO ICFR Indicator will be initiated when **all** devices on the

account have an ICFR's of < 3 dB, which simply means that there are no ICFR impairments.

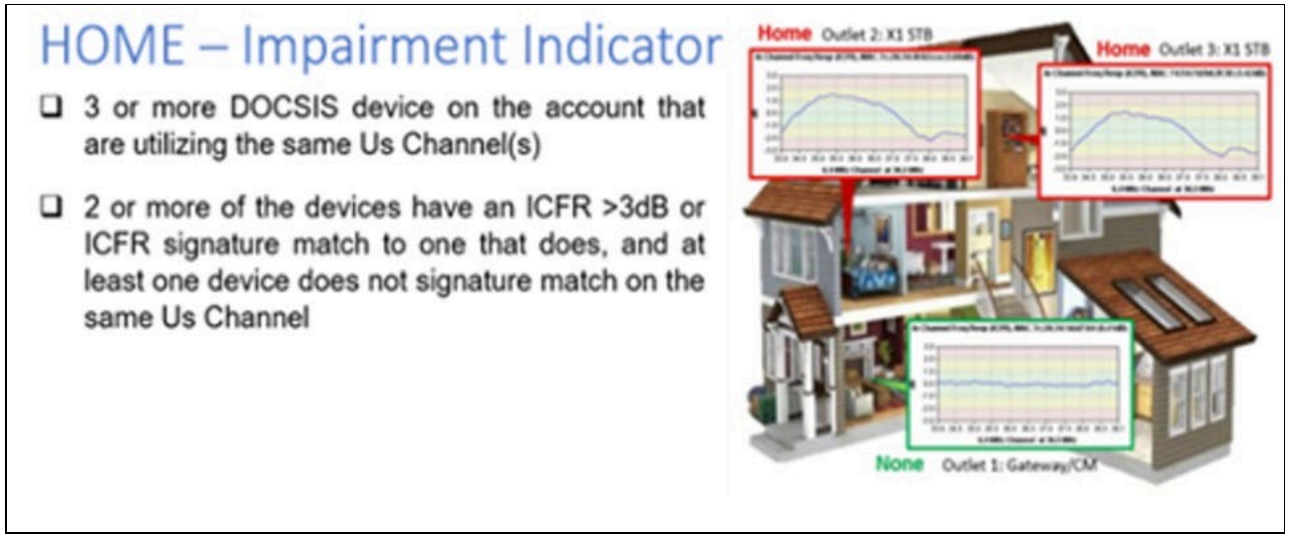
**OUTLET** ICFR impairment Indicator (Fail with Referral)

- When an ICFR fault is isolated to a specific outlet that feeds the device the device in the home; the fault is unique to the device outlet.



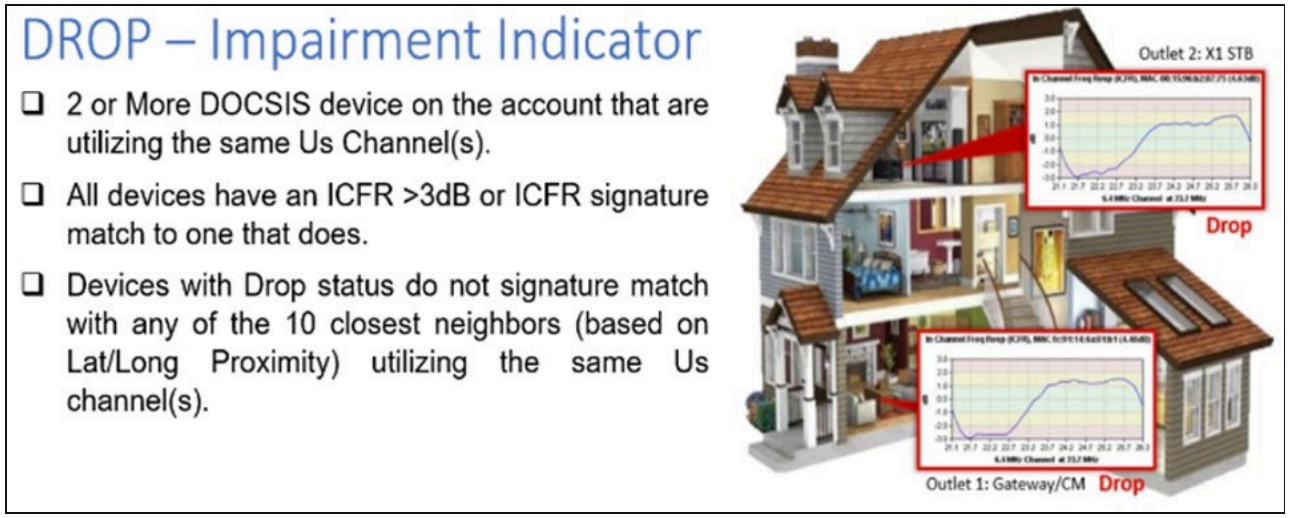

**HOME** ICFR impairment Indicator (Fail with referral)

- When an ICFR fault is isolated to a common point in the home that feeds multiple devices but does not affect all devices in the home.




**DROP** ICFR impairment Indicator (Fail with referral)

- When an ICFR fault is isolated to a common point that feeds all devices in the home. We have checked neighbors and eliminated a potential network issue as the cause.




**SOLO** ICFR impairment Indicators (Fail with referral)

- When we cannot isolate the fault to specific location in the home or drop, but we have checked neighbors and eliminated a potential network issue as the cause.

## SOLO – Impairment Indicator

- ❑ Only one DOCSIS capable device on account with ICFR >3dB or account with multiple devices with ICFR >3dB that cannot be compared due to:
  - D2 devices not utilizing the same Us channel(s)
  - Devices that are unresponsive or do not return all Us equalization coefficients
- ❑ Devices with SOLO status do not signature match with any of the 10 closest neighbors (based on Lat/Long Proximity) utilizing the same Us channel(s




**NETWORK** ICFR impairment Indicator (Pass w/ Override)

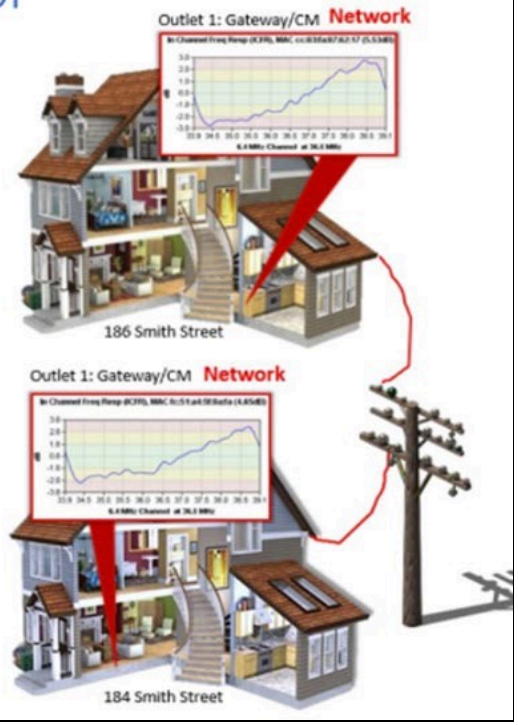
- When an ICFR fault is isolated to a common point in the network that feeds multiple accounts. At least one other neighbor account has an ICFR signature match.

## NETWORK – Impairment Indicator

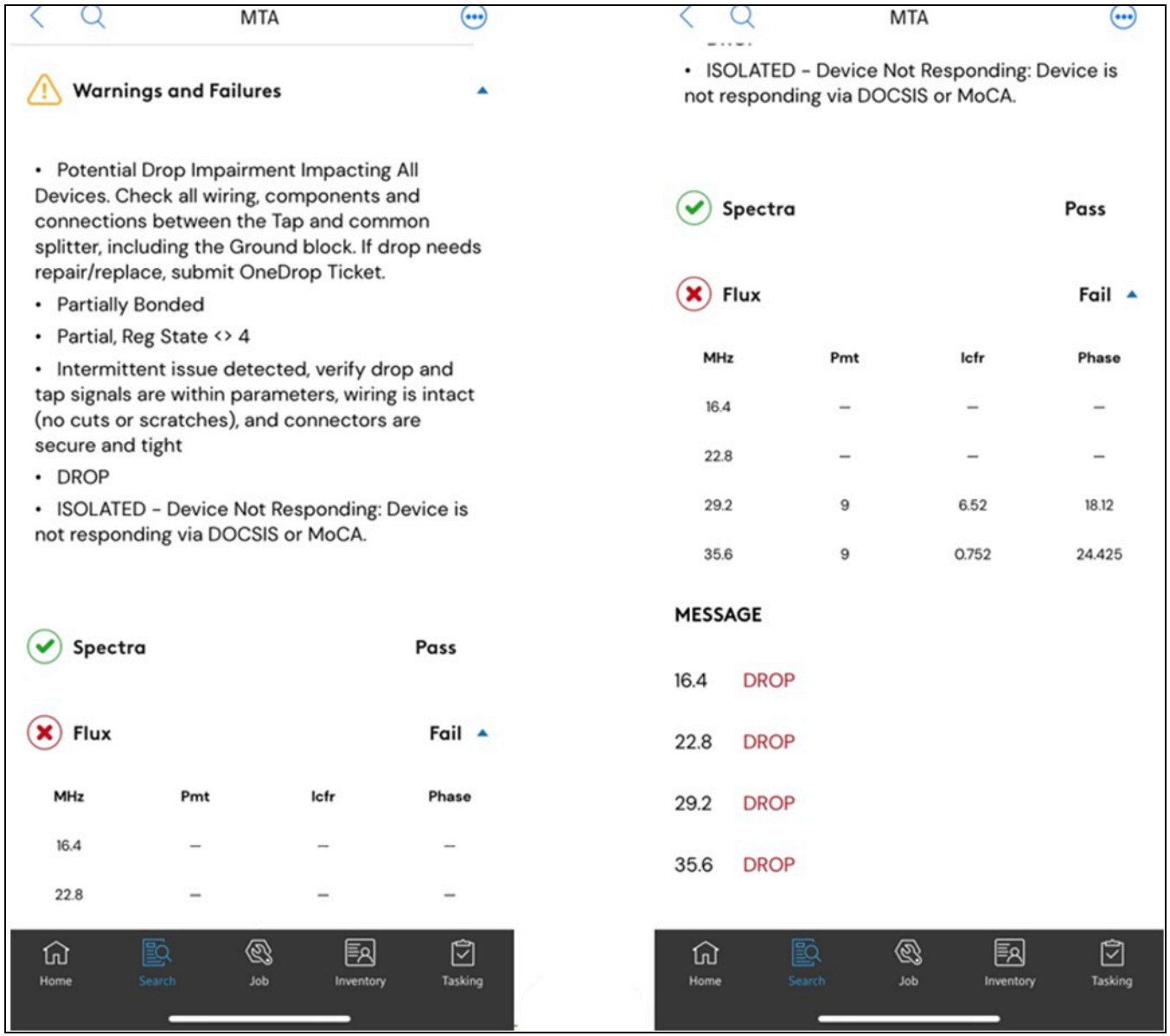
- ❑ Devices on at least 2 of the 10 neighboring accounts checked have an ICFR >3dB or ICFR signature match to one that does.
- ❑ Neighbors are the 10 closest accounts (based on Lat/Long Proximity) utilizing the same Us channel(s).

## Tech Tips

- ❑ If we get a NETWORK indicator triggered on our account/device results, the ICFR impairment impacting the device on our account has been checked against other neighboring devices and signature matches with them
- ❑ This is telling us that there is highly likelihood this ICFR impairment is part of a lager network related issue. Therefore, no RTM submissions are required unless customer has another service affecting issue with other signal levels threshold feeding the home



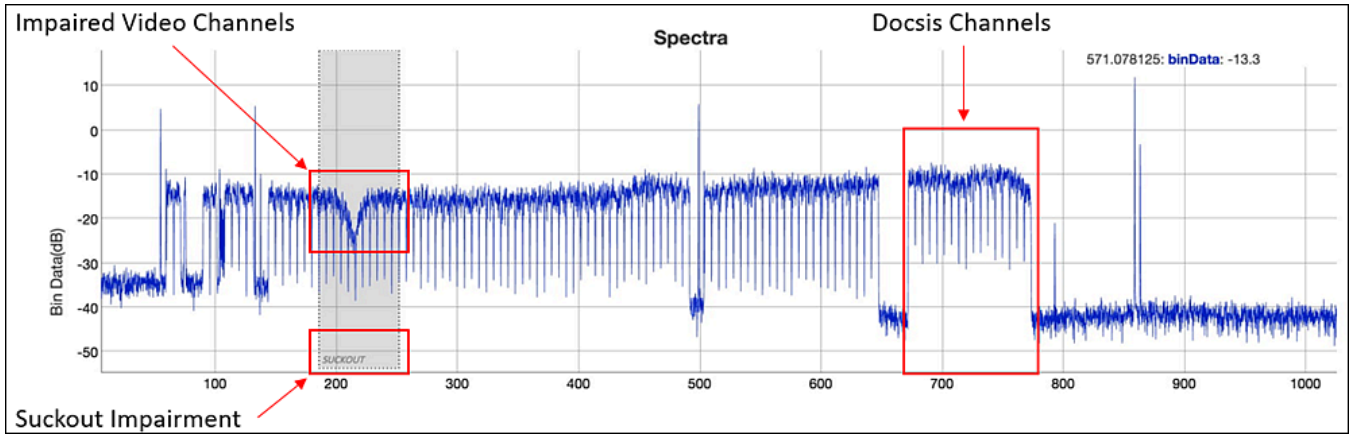
The enhanced **ICFR Indicator** logic automates the detection of specific impairments that are highly indicative of potential Flux cavities and/or shielding integrity issues. Then applies advanced correlation logic to determine if it is one of the indicators described (Outlet, Home, Drop, Solo, and Network).



## Full Band Capture (FBC) Spectrum impairments

Until recently, we have solely relied on our DOCSIS channels to infer the DS performance of our network. Now, with FBC capabilities available on almost 50% of our DOCSIS devices, we have an unprecedented view of the entire DS spectrum at the input of our devices. This has given us the ability to identify frequency specific issues in the DS spectrum that we previously could not see with our traditional capabilities.

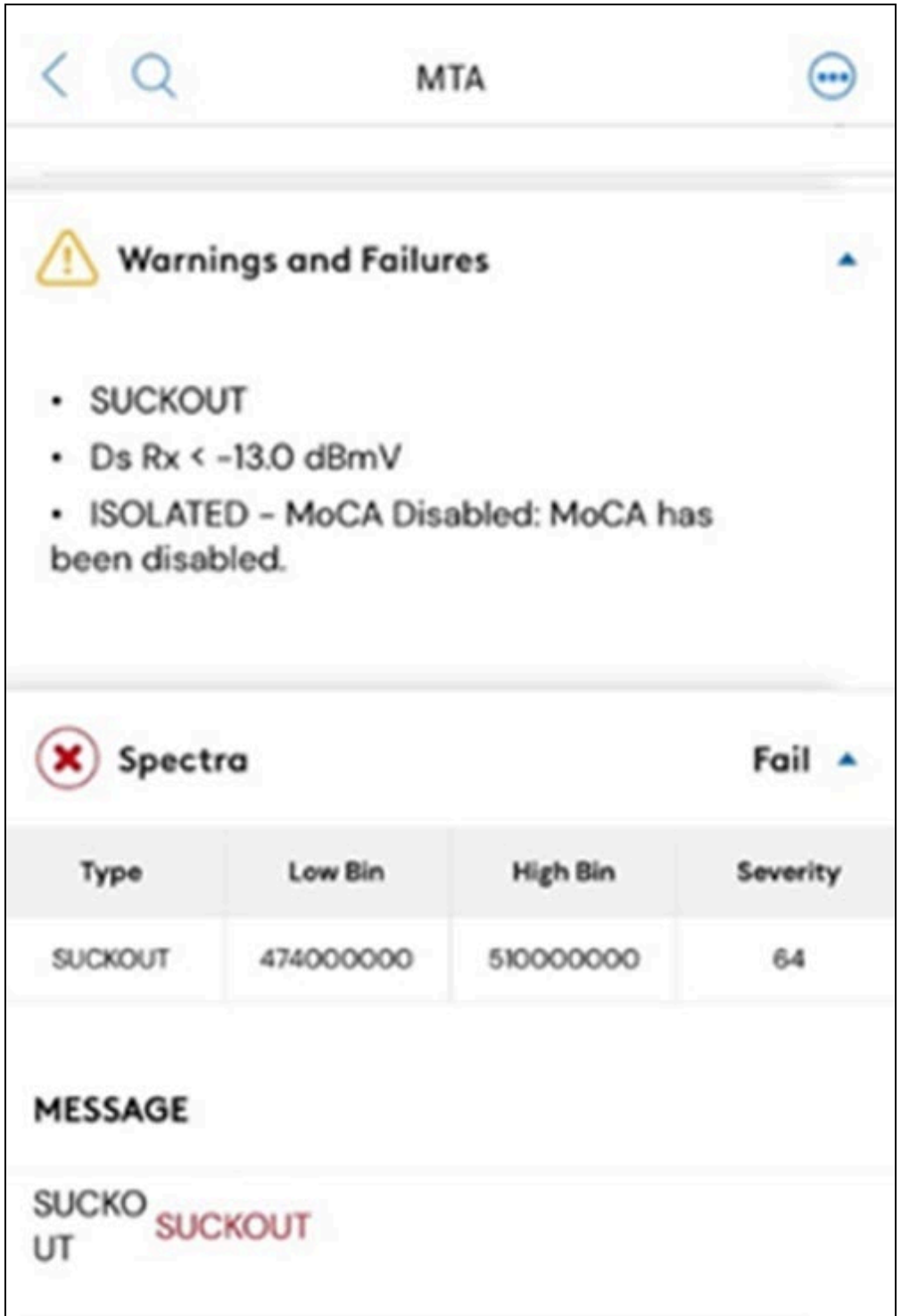
Many of these impairments are caused by issues in the customer premise and/or soft drop cabling and can be localized to the customer premise. Any impairment, other than Adjacent Channel Power, detected on the live poll of FBC capable devices will result in test failure if not part of a known HFC network event in NWT.

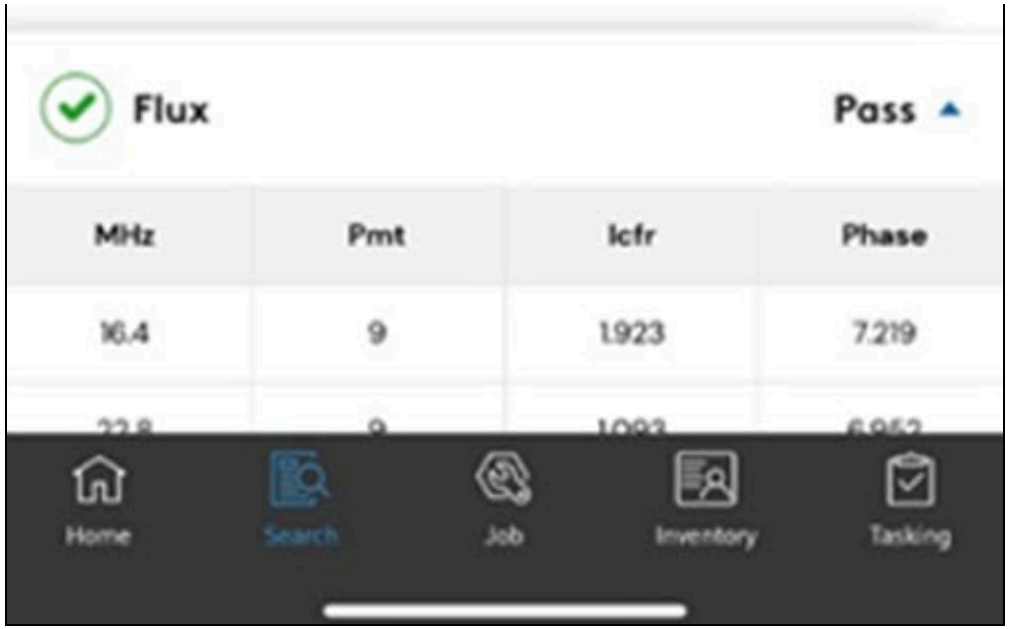



**Sample of Spectrum graph (not all regions may show DOCSIS channels in same location)**

This is not a reflection of true DS RX levels (about 15 dB lower), this an algorithm based view. Technicians should get a similar view (spectrum view compared to full scan view) when running full scan option on their meter behind Full Band Capable Device.

**Tech360 device details example of the Spectrum “Suck Out” warning message**


MTA
Warnings and Failures
• SUCKOUT
• Ds Rx < -13.0 dBmV
• ISOLATED - MoCA Disabled: MoCA has been disabled.
Spectra
Fail
Type
Low Bin
High Bin
Severity
SUCKOUT
474000000
510000000
64
MESSAGE
SUCKOUT
SUCKOUT




## MoCA PHY rates

MoCA allows us to offer multi-room services such as AnyRoom DVR. MoCA PHY rate is the actual physical layer bit rate (Mbps) of data transmitted between nodes in a MoCA network. A lower MoCA PHY rate is an indicator of subpar performance and can be caused by high path loss, home amplifiers, splitters, etc.

Accounts that have MoCA services will be tested for PHY rate performance. Any PHY rates below 200 Mbps will result in a test failure. This will be any gateway to Client scoring. Client (Xi) to Client (Xi) will not be scored but shown.



MoCA Result: Success

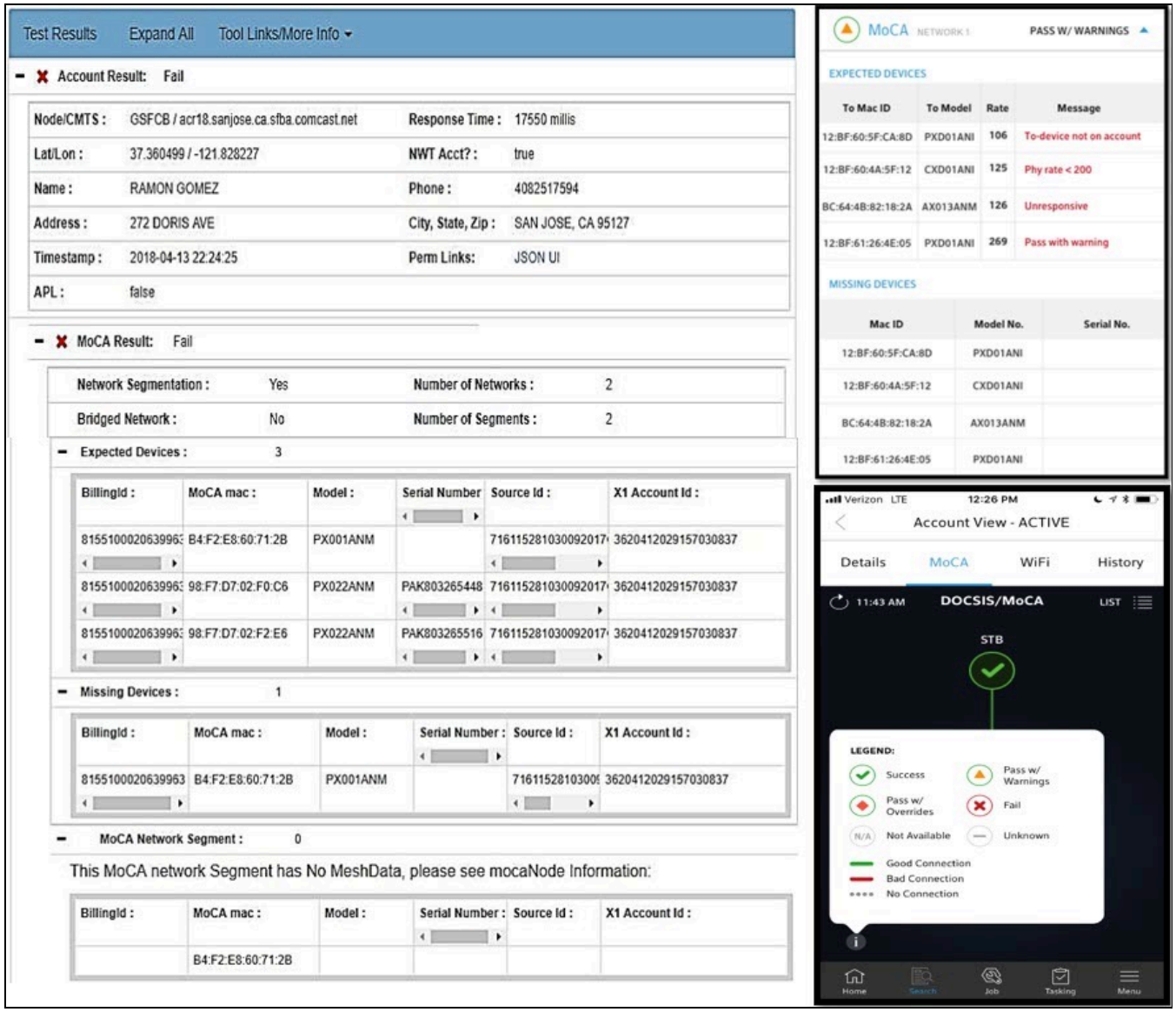
| | Idx | From Mac | From Model | From Name | Idx | To Mac | To Model | To Name | Rate | Message |
|---|---|---|---|---|---|---|---|---|---|---|
| ✓ | 0 | 70:85:C6:D6:64:EB | | | 1 | 10:86:8C:E2:50:17 | | | 253 | |
| ✓ | 0 | 70:85:C6:D6:64:EB | | | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Ro... | 232 | |
| ✓ | 0 | 70:85:C6:D6:64:EB | | | 3 | 84:E0:58:D8:98:42 | PX032ANI | Jaxon | 266 | |
| ✓ | 0 | 70:85:C6:D6:64:EB | | | 4 | 3C:DF:A9:23:1D:A6 | MX011ANC | Master B... | 271 | |
| ✓ | 0 | 70:85:C6:D6:64:EB | | | 5 | 98:1D:FA:B6:BB:66 | SX022ANC | XG2 | 266 | |
| ✓ | 1 | 10:86:8C:E2:50:17 | | | 0 | 70:85:C6:D6:64:EB | | | 267 | |
| ✓ | 1 | 10:86:8C:E2:50:17 | | | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Ro... | 225 | |
| ✓ | 1 | 10:86:8C:E2:50:17 | | | 3 | 84:E0:58:D8:98:42 | PX032ANI | Jaxon | 267 | |
| ✓ | 1 | 10:86:8C:E2:50:17 | | | 4 | 3C:DF:A9:23:1D:A6 | MX011ANC | Master B... | 266 | |
| ✓ | 1 | 10:86:8C:E2:50:17 | | | 5 | 98:1D:FA:B6:BB:66 | SX022ANC | XG2 | 265 | |
| ✓ | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Room | 0 | 70:85:C6:D6:64:EB | | | 240 | |
| ✓ | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Room | 1 | 10:86:8C:E2:50:17 | | | 231 | |
| ✓ | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Room | 3 | 84:E0:58:D8:98:42 | PX032ANI | Jaxon | 229 | |
| ✓ | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Room | 4 | 3C:DF:A9:23:1D:A6 | MX011ANC | Master B... | 236 | |
| ✓ | 2 | E0:B7:B1:09:B8:45 | PX001ANC | Living Room | 5 | 98:1D:FA:B6:BB:66 | SX022ANC | XG2 | 233 | |
| ✓ | 3 | 84:E0:58:D8:98:42 | PX032ANI | Jaxon | 0 | 70:85:C6:D6:64:EB | | | 264 | |

## MoCA (Segmented network, bridged network, and expected devices)

### MoCA Segmentation will Fail with Hard-Stop; Unexpected/Foreign Devices will Pass with Warnings

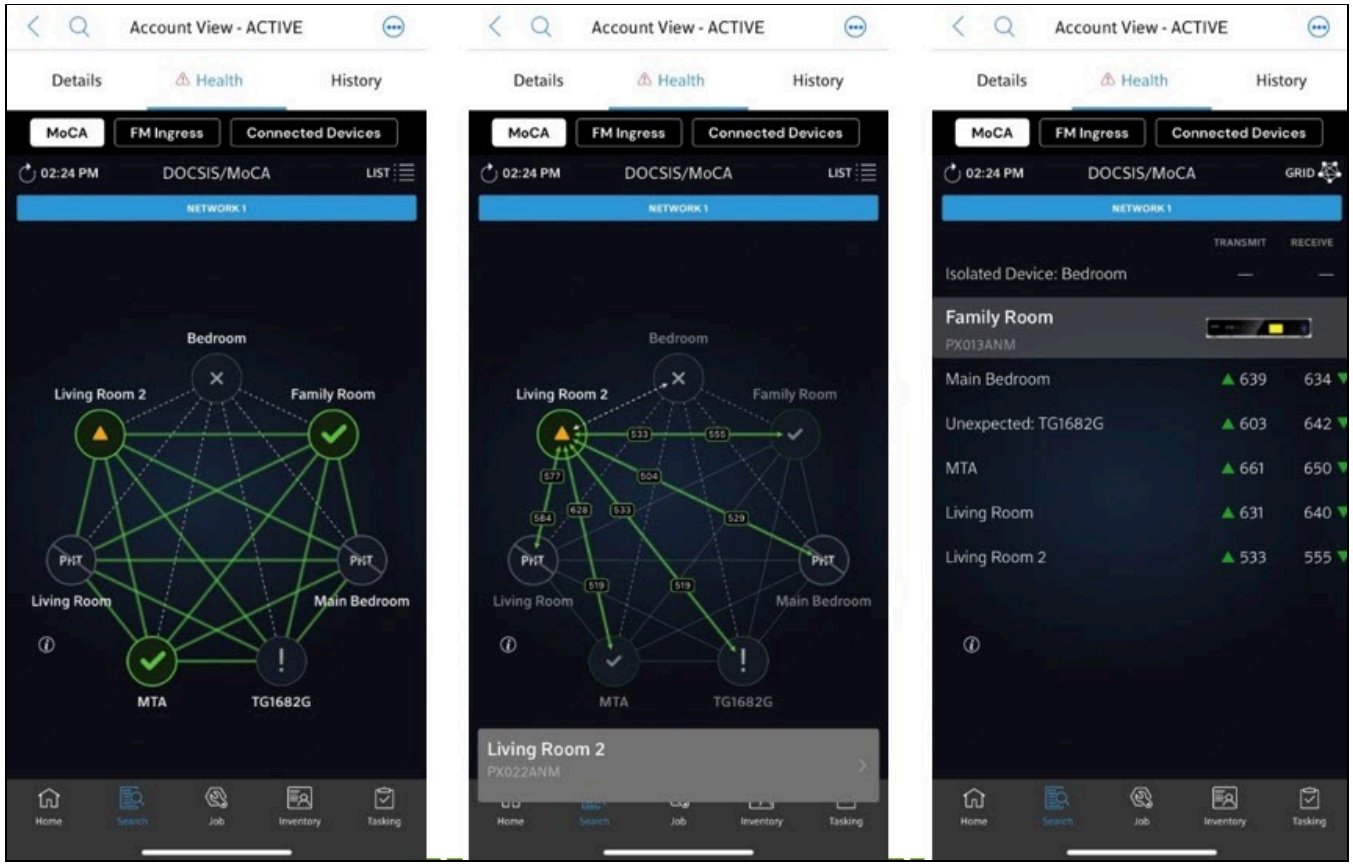
In the example below, in addition to displaying that the customer is failing for a Segmented MoCA Network (along with Bridged Networks are now considered failing conditions in PHT).

This screenshot also shows you the other newly added element included in the MoCA display, which are the **Expected Devices**, **Missing Devices**, and the details for each of the MoCA segments detected on this account.



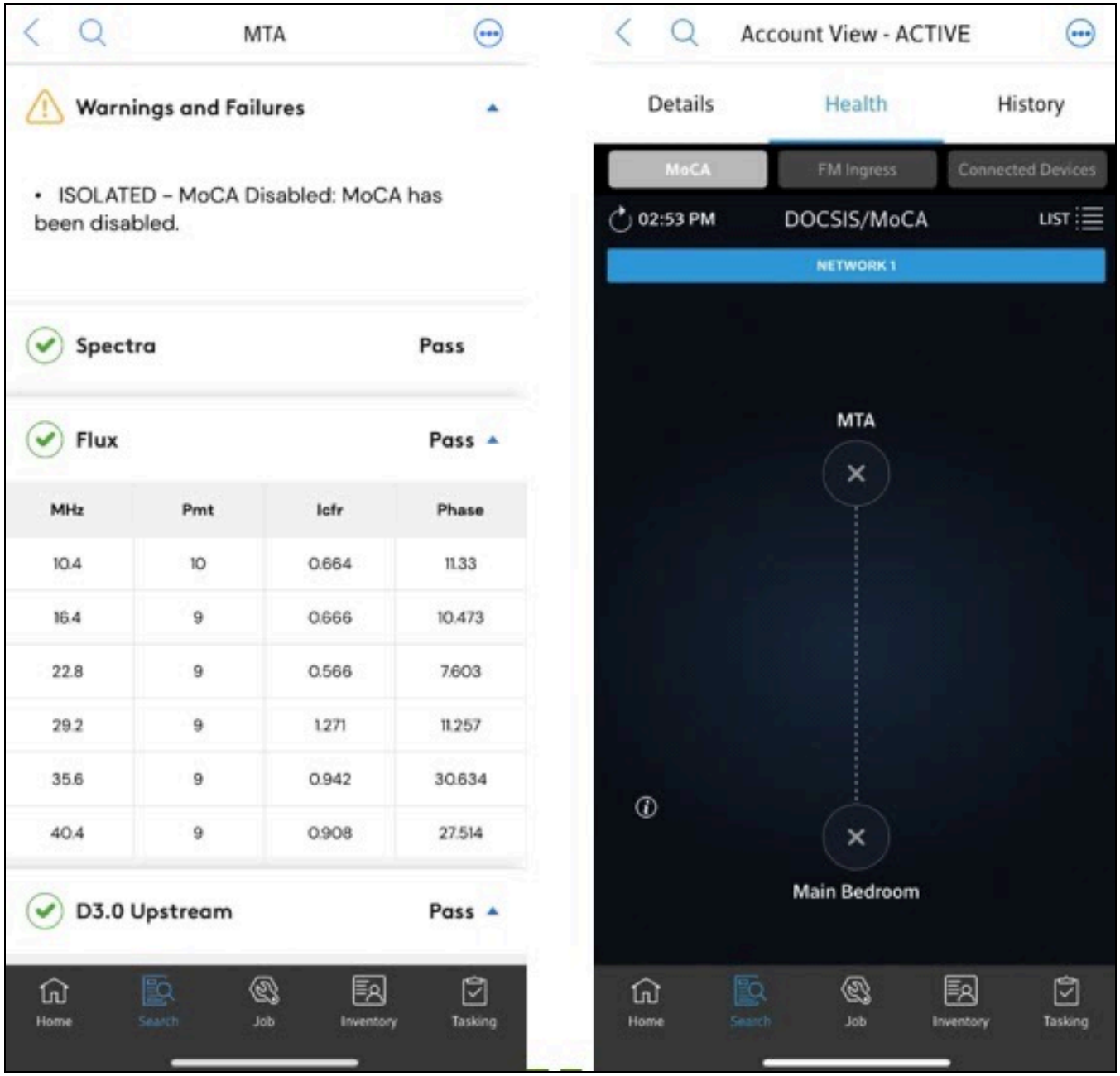

**Segmented network:**

Refers to a MoCA network that is segmented due devices that are missing from the number of devices that are expected to be on the account. Ask the customer about missing devices to see if you can resolve.




**Bridged network:**

Refers to accounts with MoCA devices from two different accounts communicating on the same network. Primary resolution is to install a MoCA POE to keep different MoCA network from bridging one another.


MTA
Warnings and Failures
• ISOLATED – MoCA Disabled: MoCA has been disabled.
Spectra
Pass
Flux
Pass
MHz
Pmt
Icfr
Phase
10.4
10
0.664
11.33
16.4
9
0.666
10.473
22.8
9
0.566
7.603
29.2
9
1.271
11.257
35.6
9
0.942
30.634
40.4
9
0.908
27.514
D3.0 Upstream
Pass
Home
Search
Job
Inventory
Tasking
Account View - ACTIVE
Details
Health
History
MoCA
FM Ingress
Connected Devices
02:53 PM
DOCSIS/MoCA
LIST
NETWORK 1
MTA
Main Bedroom
Home
Search
Job
Inventory
Tasking

## FM ingress

**FM ingress for severe noise will fail — No hard-stop; Moderate noise will pass**

In the example below for FM Ingress, you will see a new checkbox (similar to the way you would check to include Spectrum, MoCA, etc. in your PHT test) that is normally not checked by default. PHT test will validate FM Ingress.

With the inclusion of FM Ingress, you will see a new section showing up under each of the MACs on the account. This section will indicate if there is 'Severe' or 'Moderate' FM Noise conditions detected on the account; Severe will result in failing conditions and need to be resolved while Moderate will result in passing condition.

There needs to be five spikes in the FM band (88-108MHz) that are at least =>10dB above the average amplitude of that area for moderate or =>15dB for severe.

- There needs to be five spikes or more in the FM band (88-108MHz) peaking above a specific threshold.
- If these five or more spikes are at least =>10dB above the average amplitude for that given frequency area, then that condition will trigger a MODERATE FM Ingress condition. Resulting in a Pass.
- If you are in a Moderate state and any of your ingress spikes start to peak =>15dB above the average amplitude, then that condition will trigger a SEVERE FM Ingress condition. Resulting in a failure.

Our drop system is designed to be a closed loop and signal tight system; so keep that in mind when troubleshooting in-home coaxial drop system. Make sure we don't have any breaks in the coax and all connectors are signal tight.

- Ensure all connectors are corrosion free, tight, and waterproofed, all the way out to the drop connections.
- Check that cables are not corroded; do not bear signs of wear with breaks, throughout the entire drop system.

- Check the fittings throughout the home; start with connectors at the failing device and then work back from there.
- Ensure that wiring in the homes meets Comcast shielding specifications.




**Downstream FM ingress**

Service affecting examples

- Freezing and Digital Tiling,
- One Moment Please,
- Slow Download Speed,
- Latency, Packet Loss,
- Dropped Calls

- Call Quality Issues,

## EPON

**Ethernet Passive Optical Network (EPON) device outside of threshold will fail; with override options**

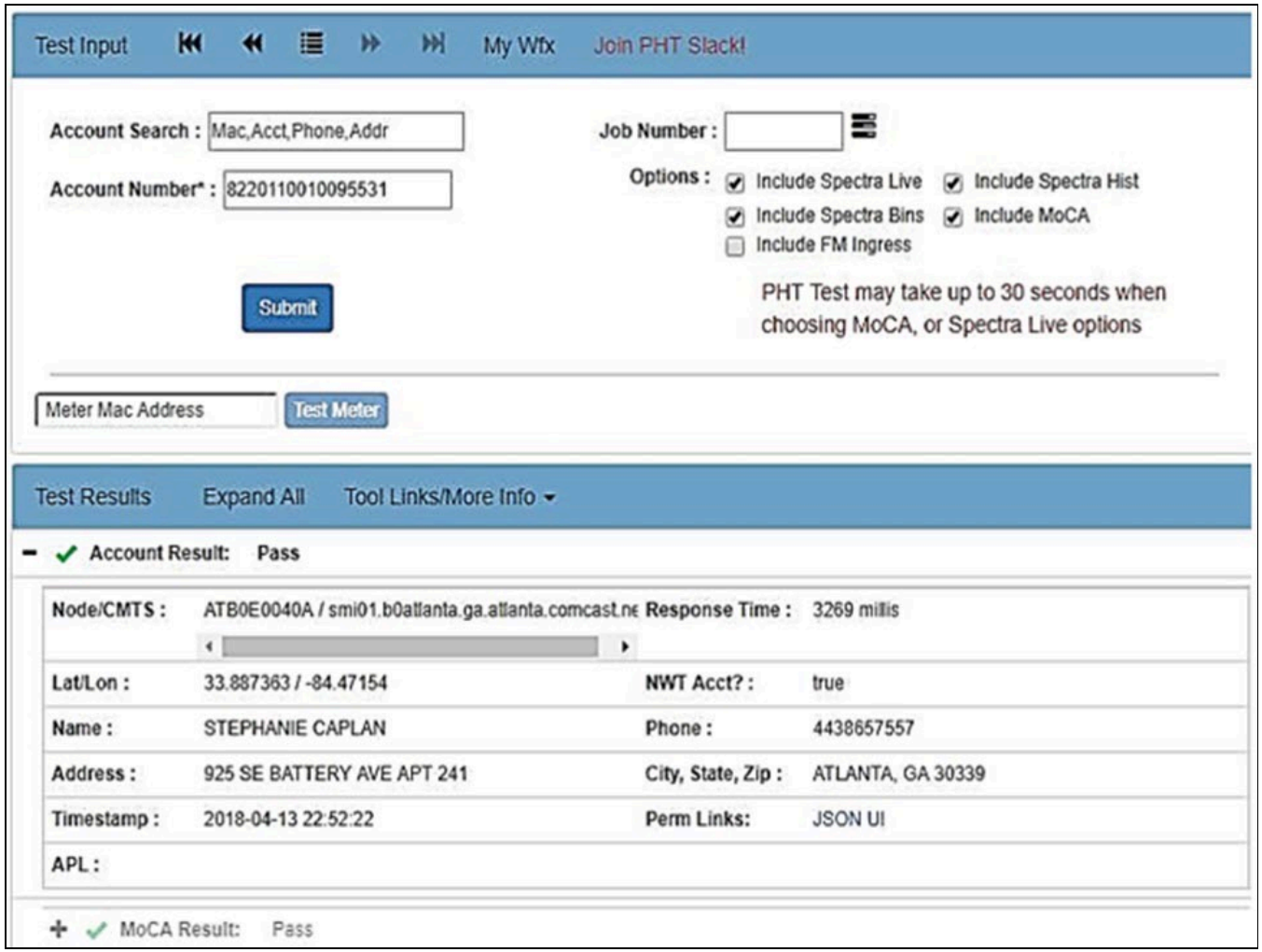
The last enhancement gives us visibility into EPON devices. We had to re-design the PHT UI to correctly reflect the data associated with an EPON account with an XF3 Advanced Fiber Gateway. The following light level thresholds will define the pass/fail criteria for these accounts. Please refer to your local referral process when any of these light levels outside of the thresholds listed below may result in a failure.

**EPON Pass/Fail thresholds (power light levels)**

Us Rx (vBng only) >= - 28.0 dBm to <= - 6 dBm

Ds Rx >= - 28.5 dBm to <= - 10.0 dBM

Below is an example of what an EPON test result will look like in PHT



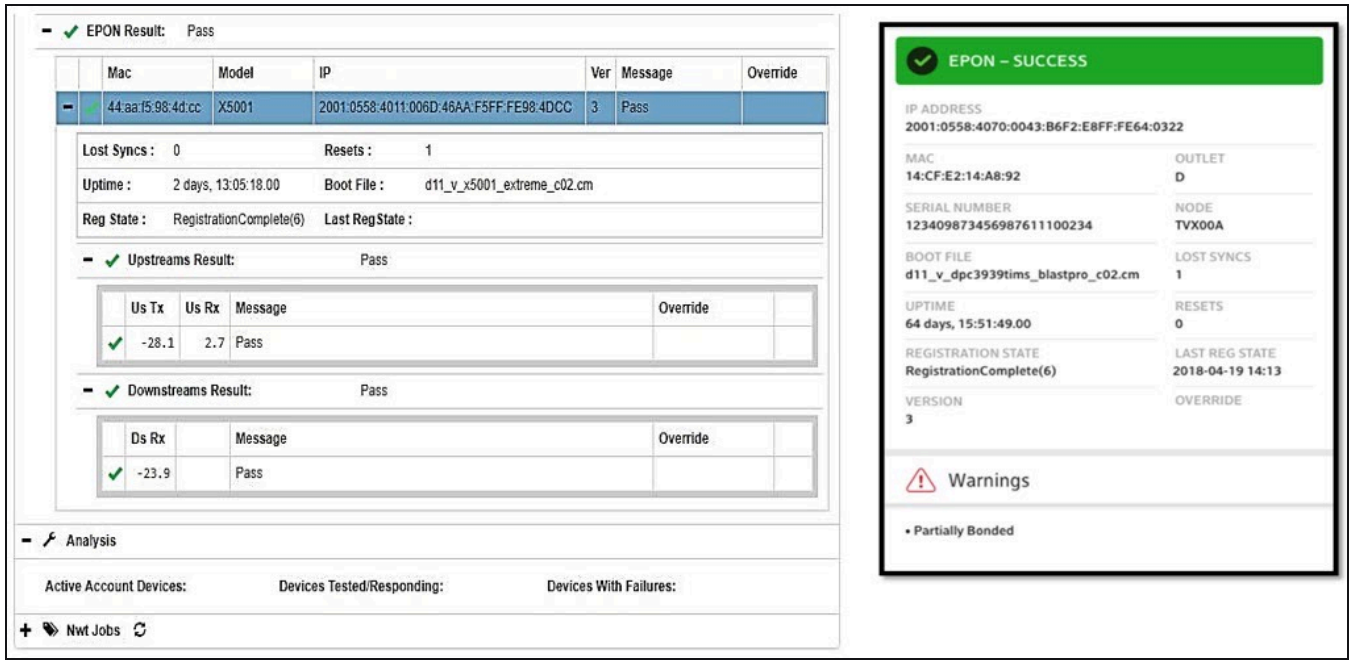

### Light levels influenced by

- Dirty fiber connector
- Poorly spliced connector
- Scored/damaged fiber
- Incorrect settings on meter

### Fiber optic tips and reminder

- Always inspect with fiber scope

- Inspect all ports (XF3, Bulk-Heads, Fiber NAP/MST, etc.)
- Clean connector and/or ports (only if needed)
- Check Power Light Levels
- Re-inspect with fiber scope
- Connect fiber right away after passing scope test acquired




### WiFi RSSI

RSSI Validation will be a Pass or Fail Threshold; <= -70 dBm is a Pass and > -70dBm is a Fail

With the PHT v6 on May 7, 2019; we will now be able to validate connectivity within a customer's premise on all three fronts; DOCSIS, MoCA, and WiFi-RSSI signal connectivity. The first phase of WiFi-RSSI validation will focus on the wireless connectivity between Xi5/Xi6 set-top receivers and our XB gateways in the customer home.

**Note:** We will not have visibility and validation capabilities of our XFi Pods or any Xi5/Xi6 set-top receivers. This validation capability will come in a future release of PHT enhancements.

### RSSI pass/fail threshold

- Pass (Greater Than or Equal To -70 dBm) >= -70 dBm
- Fail (Less Than -70 dBm) < -70 dBm



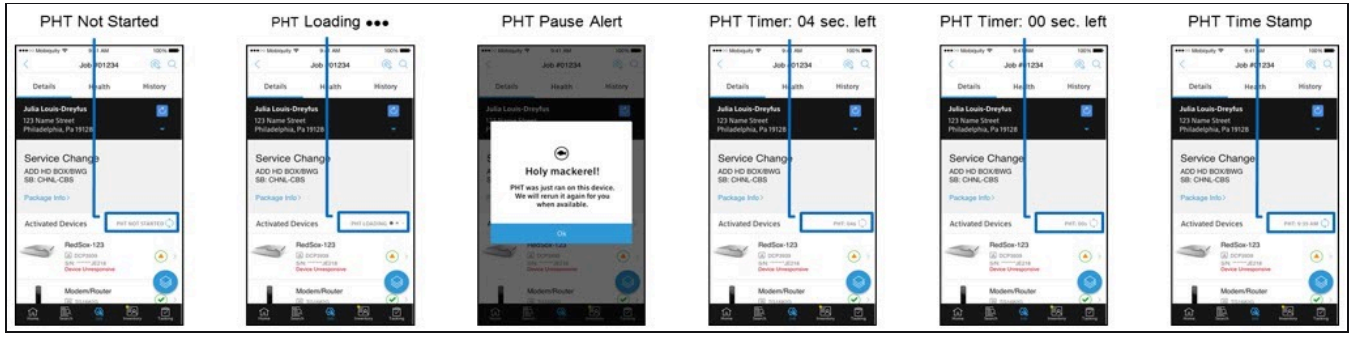

## PHT Pause (Test throttling)

PHT will throttle the number of tests that can be requested per account number. An account can be tested once every 90 seconds. The purpose is to assist with unresponsive responses rates that occur when a DOCSIS device on an account is polled too many times in within seconds of one another.

- We will use the latest time stamp from JSON to invoke a timer to show the "remaining" time that is left in the pause based on a 90 second timeline (e.g. JSON shows request was made 37 seconds ago; the timer will show 53 seconds left in PHT Pause)
- Users will be given a Buzz alert that they have been Paused from initiating another test because one is in progress.

- **Note**: If the remaining pause time is less than three seconds, then user will be allowed to retry, with no buzz alert stopping them.

The screenshots below give us examples of some of the proposed PHT Pause messaging a technician may see.




## Unresponsive device

CMTS knows the device has been online within the prior 24 hours but device offline; PHT will Pass with a warning that device for being unresponsive due to reasons out of your control.

What should technician do?

- Ensure the device is supposed to be on the account. On installs/Change of service, ensure the device matches what billing has for the customer.
- Ensure device is powered on; refrain from polling the devices in a rapid succession, i.e. within 90 seconds of each other. Try a UDB/XBO refresh on the account if an X1 or an XB3/6 is being swapped.




PHT informational data:

| PHT informational data | |
|---|---|
| **Data** | **Description** |
| Incomplete D3P data | One or more devices on the account did not return all requested data needed for PHT |
| Service call failure | A call to any service used to gather PHT data failed for the account tested |

**Notes:**

- Unresponsive device = offline < 24 hours
- Unrecognized device = offline > 24 hours

## Unrecognizable device

Active device on an account (account at least 3 days old) that has been unresponsive for longer than 24 hours.

What should a technician do?

- Validate with customer that device is still in the home and why it may be offline (sparingly used device, etc.).
- May need to be re-run due to failure in connectivity. The device may be in an “Active” status and should be set to “disabled” status.

- In addition, technicians should ensure there isn't a placeholder on the outlet causing PHT to look for the device.

## Incomplete D3P data

Poll for data is missing data (could result from poor cell connectivity or disruption during test).

What should a technician do?

- Ensure solid connectivity to Internet (4G/LTE or Xfinity Wifi Connection), power cycle-affected device, re-run PHT.
- Does not mean a failure to the technician, just missing data.



Upstreams Result: Pass w Warnings

| | Freq | Tx Pwr | Rx Pwr | Act. Pwr | Cm SNR | Port SNR | T3 | T4 | Reg St | Message | Override |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ✓ | 18.9 | | 0.2 | | 31.2 | 31.0 | | | | Missing Upstream D3P Data | |
| ✓ | 23.7 | | 0.0 | | 32.5 | 32.5 | | | | Missing Upstream D3P Data | |

## Service call failure

Failure to call data sources (cell connectivity, VPN drop, tools outage).

What should a technician do?

- Ensure solid connectivity to Internet (4G/LTE or Xfinity Wifi connection if available), re-run PHT.
- This does not mean a failure to the technician, the test simply failed to pull down data.



| Warnings | Warnings | Warnings |
|---|---|---|
| • Desc : Call to Iris CMTS Lookup failed | • Desc : Call to Spectra failed | • Desc : Call to MoCA failed |

## Missing PHY rate message — Call from X-Ray failed to return PHY rates for all devices

Technicians can re-run PHT. The original call may have dropped information on return due to connectivity concerns. This could also be caused by a recent swap of equipment and similar to unresponsive device, a device in ACTIVE status instead of disabled status. An account refresh may also help solve this concern.

PHT Pass w/ Override examples:

| PHT Pass w/ Overrides | |
|---|---|
| **Override type** | **Override criteria** |
| US TX | In order for an override in PHT to occur the account tested must fail and be listed as an affected account for an ongoing NMT event that matches the exact failure reason.<br><br>For example, if the account fails for a low DS SNR and it is listed as an affected account in the current NWT event for DS SNR, then the DS SNR failure will be overridden. If there are other failure reasons on the account that are not apart of a NWT event |
| DS SNR | |
| DS RX | |
| Spectra Suckout | |
| Spectra Wave | |

**Note:** USSNR does not trigger a Pass w/ Override condition. Technicians should work with their supervisor and local maintenance team to verify if an US SNR condition is a Service Affecting. A Pass w/ Override does not automatically need an RTM for any other metric, override has already captured the event and is being prioritized for work.

## PHT analysis types and descriptions — These analysis are given when data suggests one of these conditions could exist

| PHT analysis types | |
|---|---|
| Analysis type | Analysis description |
| Filter | Based on the US registration states of one or more devices is potential noise mitigation file present. |
| Outage | None of the expected DOCSIS capable devices on the account are responding and NWT has an outage event in progress that the tested account is affected by. |
| POE filter | There are devices not listed on the tested account in billing that are communicating on the tested account's MoCA network indicating a missing POE filter. |

| PHT analysis types | |
|---|---|
| SNR Delta | The CM US SNR on a tested device is more than 5 dB lower than the port level US SNR indicating a potential device or in-home issue. |
| True node | The CMTS and/or US interfaces being used by devices on the tested account do not match the billing and/or NCP listings indicating a billing or NCP error. |

## Filter — Possible noise filter blocking one of the US from connecting

Technicians need to check for any potential noise filters at the tap and ensure the home is free of ingress prior to removing the trap. After making the corrections, technicians should re-run PHT to get results.



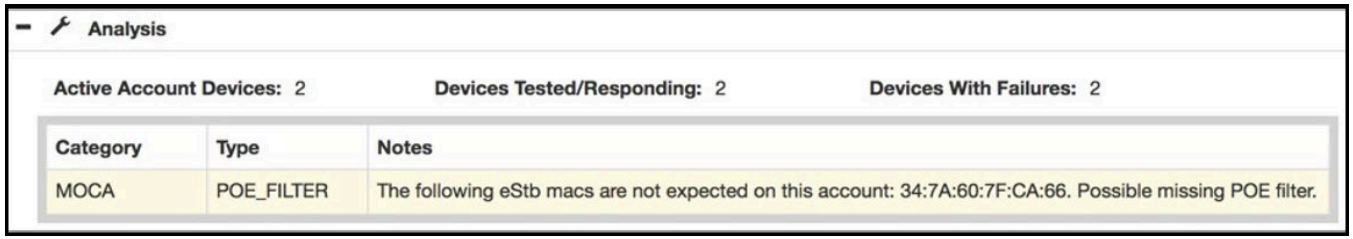

## Outage — All devices offline and matching outage in NWT

Technicians should confirm the outage with supervisor/XOC and inform customer of estimated ETA of the correction (if one is available).



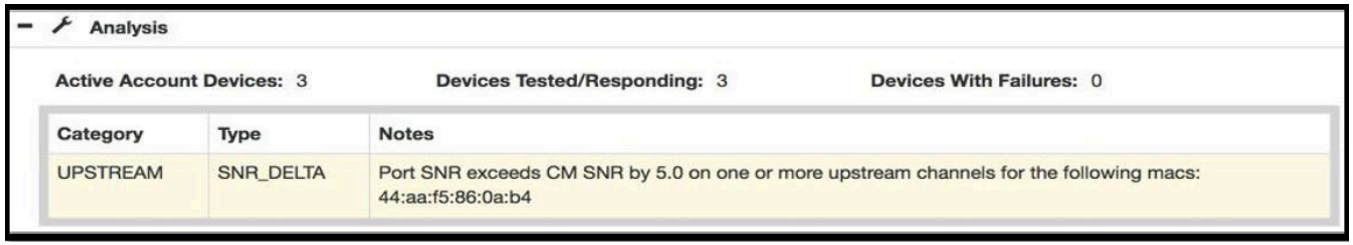

## POE filter — When another device is detected inside a MoCA mesh network that is not on the account (neighbor device likely)

Technicians need to check for location of the POE filter within the home, may require a relocation of the POE filter to input of the first splitter in an MDU setting (most common occurrence of messaging). Re-run PHT after work is performed to get result. If it is a transfer, ensure that a UDB/XBO refresh is sent to sync the accounts up. (Available in XRAY or TECH360.)



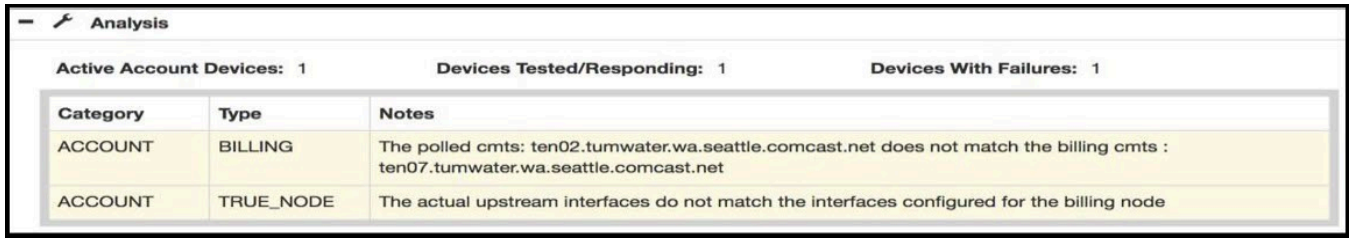

## SNR Delta — Delta between Port SNR at CMTS and cable modem SNR

Technicians need to investigate why the SNR of the cable modem of the home is degraded from CMTS SNR. A delta of more than 5 dB indicates potential RF impairments in the home.




True Node- Device is on a CMTS that does not match the billing node

Technicians should follow local guidance on correcting these accounts, may require escalation to XOC or Construction teams within the region to correct the mismatch. Consult your supervisor on best direction for your region division.




## ALL OUT — All devices offline and unresponsive; No matching NWT event or outage

If there are two or more devices in the home and all are unresponsive and showing offline, you will now get a PHT notification that we have an All-Out condition. This should prompt a technician to start checking the signal feeding the customer's in-home network.

## PHT failures and NWT node event override

There are some cases where PHT will fail because certain metrics are out of threshold due to a HFC network issue. During the PHT test, we also pull in all open NWT events on the customer's node.

- If any metric fails, we cross reference it with the open HFC events.
- If the event is open for the metric that fails, we then make sure that the customer is part of or in very close proximity to the event.
- If all criteria is met, we will override the failure and pass the technician's test.

## Troubleshooting

***How do I correct an unrecognizable device?***

This is a device that has been offline longer than 24 hours but is still showing up in billing. Ensure that the device actually is in the home or being used by the customer. Working with your supervisor or the TSC the device should be moved to a disabled status so it does not show as an active device.

***How do I correct an unresponsive device?***

This is a device that is expected to be online but is showing offline.

- Ensure that the device is supposed to be on the account. In a new install/Change of service, ensure the devices match what billing has for the customer.
- Ensure all devices are powered on and not unplugged.
- Refrain from polling the devices in a rapid succession. Avoid running a test or any other tools within 60 seconds of each other.
- Try a UDB/XBO Refresh on the account if it is X1 or an XB3/6 being swapped.

**Note**: In the event of extreme billing latency (i.e. swapped devices/installed devices taking a long time to be responsive), please send the examples to your supervisor to escalate or post to the #pht-National Slack channel.

| Metric | Lower Fail | Upper Fail | One Device Failure of Multiple Devices on Account | Multiple Device Failure on Account | Single Device Account Failure | RTM submittal requirements for each PHT failure type |
|---|---|---|---|---|---|---|
| Actual US TX | < 25 dBmV | > 54 dBmV | Check US TX levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and connections that feeds device. | Check US TX levels on all channels with meter or test device at customers tap, ground block, and/or the consolidated splitter that feeds all devices that failed. | Check US TX levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and connections that feeds device. | Technician must run DOCSIS test at customers tap and US TX levels must be out of spec. for the tap value in accordance with your local market specifications. Technician must provide US TX levels for all USUS channels as recorded at the customer's tap on RMT ticket. |

| | | | | | |
|---|---|---|---|---|---|
| Partial Mode/Red State | Any US channel Reg state <> 4 or DS channel Reg state <>1 | Check for potential US noise filter at both ends of the cable that feeds the failed device. Check for impairments in the DS band. If no filter/impairment is present, power cycle device and retest. | Check for US noise filter on customers drop at tap, ground block and/or consolidated splitter that feeds all devices that failed. If no filter is present, power cycle devices and retest. | Power cycle device and retest. If device still in partial mode check for US noise filter at both ends of the cable that feeds the failed device. | Technician must attempt to lock meter in DOCSIS 3.0 more or DOCDID 3.0 test device directly at the customer's tap and the meter/test device must lock in partial mode and not achieve fill US bonding. Meter or test device must be used to ensure CPE is not the issue. Technician must provide specific channels that are in partial mode as recorded at the customer's tap on RTM ticket.x |

| | | | | | |
|---|---|---|---|---|---|
| Flux ICFR | ICFR > = 3 dB and PMT = tap 9 | Verify the integrity of the cable and all connections that feed the failed device. Run test lead to device from consolidation splitter and retest. | Run test drop to customer's device from the tap port or ground block and retest. If still bad with test line at the ground block test directly from tap port, if bad directly from tap port submit RTM, if good at tap troubleshooting drop issue. | Verify the integrity of the cable and all connections that feed the failed device. Run test lead to device from consolidation splitter and retest. | Technician must run a test leada from the csutomer's tap directly to customers device and/or retest PHT. Technician must provide ICFR and PMT values for all US channels as recorded at the customer's tap. ICFR directly from the tap should also fail PHT in order to submit an RTM for this metric. |

| | | | | | | |
|---|---|---|---|---|---|---|
| DS RX | < -13 dBmV | 13 dBmV | Check DS RX levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and all connections that feeds device. | Check DS RX levels on all channels with meter or test device at customers tap, ground block, and/or the consolidated splitter that feeds all devices that failed. | Check DS RX levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and all connections that feeds device. | Technician must check the DS RX levels for all DOCSIS channels at the customer's tap and DS RX levels must be out of spec. for the tap value in accordance with your local market specifications. |
| DS SNR | < 33 dB | | Check DS SNR levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and all connections that feeds device. | Check DS SNR levels on all channels with meter or test device at customers tap, ground block, and/or the consolidated splitter that feeds all devices that failed. | Check DS SNR levels on all channels with meter or test device at the point that feeds device that failed. Verify integrity of the cable and all connections that feeds device. | Technician must check the DS SNR levels for all DOCSIS channels at the customer's tap and DS RX levels must be out of spec. for the tap value in accordance with your local market specifications. |

| | | | | | |
|---|---|---|---|---|---|
| Spectra Impairments | Any detected impairment other than ACP that is not overridden by a known network level event | Run a full channel scan at both ends of outlet that feeds failed device. Verify the integrity of the cable and all connections that feed the failed device. | Run a full channel scan with your meter at customer's tap, ground block, and/or the consolidated splitter that feeds all devices that failed to isolate where impairment begins. | Run a full channel scan at both ends of outlet that feeds failed device. Verify the integrity of all cable and all connections that feed the failed device. | Technician must use meter to run a full channel scan at customer's tap and verify the impairment is visible from the tap port. Technician must provide details on the specific impairment, the channel power and MER levels of all impacted channels. |
| MoCA PHY Rate | Any link to or from network controller (never xi to xi) < 200 Mbps | Verify location of POE filter is optimal for MoCA mesh network. Verify the integrity of the cable and splitter configurations to all MoCA devices. | Verify location of POE filter is optimal for MoCA mesh network. Verify the integrity of the cable and splitter configurations to all MoCA devices. | Verify location of POE filter is optimal for MoCA mesh network. Verify the integrity of the cable and splitter configurations to all MoCA devices. | N/A Technicians cannot enter a Network Maintenance RTM for MoCA failures. |

## PHT failures and NWT node event override

There are some cases where PHT will fail because certain metrics are out of threshold due to a HFC network issue. During the PHT test, we also pull in all open NWT events on the customer's node.

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